

UNICON: An Open Dataset of Electricity, Gas and Water Consumption in a Large Multi-Campus University Setting

Harsha Moraliyage *Member, IEEE*, Nishan Mills *Member, IEEE*, Prabod Rathnayake *Member, IEEE*,
Daswin De Silva *Senior Member, IEEE*, Andrew Jennings
Centre for Data Analytics and Cognition, La Trobe University, Victoria, Australia.

Abstract— In this paper we introduce UNICON, a large-scale open dataset on UNiversity CONsumption of utilities, electricity, gas and water. This dataset is publicly released as part of La Trobe University's commitment to Net Zero Carbon Emissions by 2029, for which we are building the La Trobe Energy AI/Analytics Platform (LEAP) that leverages Artificial Intelligence (AI) and Data Analytics to analyse, predict and optimize the consumption, generation and utilization of electricity, renewables, gas and water resources. UNICON contains consumption data for La Trobe's five campuses in geographically distributed regions, across four years, 2018-2021 inclusive. This includes the COVID-19 global pandemic timeline of university shutdown and work from home measures that led to a significant decrease in the consumption of utilities. The consumption data consists of smart electricity meter readings at 15-minute granularity, gas meter readings at hourly intervals and water meter readings at 15-minute intervals. UNICON also contains weather data from the closest weather station to each campus, collected at two-speed latency of 1 minute and 10 minutes. The dataset is annotated with internal events of significance, such as energy conservation measures (ECMs) and other measurement and validation (M&V) activities conducted as part of LEAP optimization. To the best of our knowledge, this is the first large-scale, comprehensive, open dataset for the three main utilities, electricity, gas, and water consumption in a multi-campus university setting. A high granularity data dictionary and technical validation of the dataset for consumption trends, baseline modelling and forecasting are further contributions of this article that will enable interested research scientists, academics, industry practitioners, sustainability and energy consultants to experiment and evaluate their AI algorithms, models, forecasts, as well as inform the development of energy benchmarks, guidelines and much needed data-driven energy policies.

Keywords: *dataset, electricity, gas, water, consumption, baseline, forecasting, Artificial Intelligence, Analytics, university dataset*

I. INTRODUCTION

An expanding consumerist society and the rapid growth of urbanization activities has resulted in an exponential increase in the demand for basic utilities, electricity, gas and water, which is expected to double by 2050 compared to 2010 [1]. The building sector consumes more than 30% of the energy consumption demand and is responsible for emitting quarter of global greenhouse gas emissions[2][3][4]. Hence reducing the energy consumption to achieve energy sustainability is identified as a key mechanism to reduce the increasing energy demand and emissions while also addressing the challenges of a

warming planet. Technological solutions play a vital role in reducing the energy demand as the cost of energy prices continues to soar across the world[5]. Many countries are incorporating new policies on energy efficiency to support climate change action. Energy Policy Act (EPA) is an executive order passed by the US government to improve the energy efficiency in 15% of the buildings by 2015 with respect to 2003 baseline[6]. Energy Efficiency Directive is issued by the European Parliament to improve the energy efficiency of industry by 20% by 2020 in all the member countries[7]. Offices and university buildings are identified as commercial buildings which are categorized as high energy-consuming structures[2]. Hence analysing energy-consuming patterns in buildings plays a vital role in planning energy efficiency projects to achieve sustainability goals and net zero carbon emission targets. With the advancements in technology, the efficiency of energy-consuming devices and appliances has increased. Retrofitting old air conditioning and cooling devices, replacement of lighting with LED lights, and modifying the operation behaviours through energy efficiency projects have reduced the energy consumption of the buildings[8]. The measurement and Verification (M&V) process emerged as the standard methodology to estimate the efficiency of the energy projects. There are well established methodologies such as Performance Measurement and Verification Protocol (IPMVP) and the American Society of Heating, Refrigerating and Air-Conditioning Engineers' (ASHRAE) Guideline 14 to estimate the energy savings of M&V projects[9]. The effectiveness of an Energy Conservative Measure (ECM) project is evaluated by calculating the amount of energy avoided after the completion of the ECM project compared to how much it would have consumed without the ECM[8].

There is a growing need from research scientists, academics, industry practitioners, sustainability and energy consultants for publicly accessible datasets to support energy and sustainability research that lead to energy benchmarks, guidelines and data-driven energy policies [2]. Recent advancements in deep learning as well as artificial intelligence open new pathways to build complex machine learning models with high accuracy to provide solutions for real-world applications[10]. Demand forecasting, energy efficiency analysis, cost prediction, peak demand analysis, building bidding strategies, and baseline modelling can be identified as a few areas which extensively use machine learning in the energy domain[11][12][13]. Hence the availability of real-world datasets with multi-year, multi-setting

data streams is a major requirement to develop and test AI-based methodologies and frameworks that deliver actionable outcomes for energy efficiency, net zero emissions and optimised management [14].

Among the existing open energy datasets, a majority are collected from the United States; Dataport[15], BLUED[16], Canada; HUE[17], RAE[18], Europe; BLOND[19], COOLL[20] and Asia; IAW[21], ENERTALK[22]. Most of the existing datasets do not contain weather data which has a direct impact on energy consumption[2]. Weather data associated with energy readings is critical to build accurate energy prediction and baseline models. Another limitation is that there are very few annotated datasets with ECM events which are invaluable to perform M&V analysis [3][6]. According to Himeur et al[2], there are no datasets which capture energy consumption over long term multi year periods. Building a high-quality real-world dataset is a challenging task. This is also due to the nature of the operating conditions of smart meters as they can be subjected to hardware and communication failures during the data collection periods[23]. In this paper, we present UNICON dataset, which contains electricity, water, and gas consumption of buildings located at the five campuses in the La Trobe university network; Bundoora, Bendigo, Albury-Wodonga, Mildura, and Shepparton. This dataset also contains the weather data relevant to each of the campus locations. It also addresses the abovementioned limitations by containing long term data from 2018-2021 inclusive, COVID-19 period, annotated for ECM events, and all relevant weather variables for more accurate model development.

II. LEAP AND NET ZERO CARBON EMISSIONS

The La Trobe Energy AI/Analytics Platform (LEAP) is the flagship AI initiative of La Trobe University's commitment towards achieving Net Zero Carbon Emissions in all campuses by 2029. The University has committed \$75 million for achieving this ambitious goal by adopting several strategies. Firstly, decreasing the dependence on the Victorian state power grid where brown coal is the primary source of energy, accounting for about 80% of electricity generation, increasing the use of renewable energy generated by rooftop photovoltaic panels, investing in battery storage, and extensive energy consumption and generation optimisation across the university campuses using AI and Analytics. As the flagship AI initiative of La Trobe's Net Zero Program, LEAP is being designed, developed and deployed as a "Living AI Lab" project, by the Centre for Data Analytics and Cognition (CDAC), La Trobe's leading research centre for fundamental and applied AI research. LEAP can be described by its key technological innovations, an all-data lakehouse, data quality framework, AI and automation engine, and multi-stakeholder analytics dashboards. The all-data lakehouse generates a single-version-of-truth, centralised repository of internal data streams on consumption, generation, utilisation, events, and external data streams of weather, climate, tariff, and energy market metrics. The data quality framework validates all data feeds to standards and formats persisted by the AI and automation engine. It also uses AI for predictive coding during dropouts, missing and erroneous sensors reads, so that the all-data lakehouse is inclusive and complete. The AI and automation engine consists of multiple intelligent and automated functions, such as baseline prediction, short-term and long-term

consumption and generation forecasting, anomaly detection using intra-day and inter-day consumption signatures, tariff simulation, metering and bill validation. The analytics dashboards visualise energy metrics, forecasts, fluctuations, trends and outliers that can be sliced, diced, rolled up, drilled down by dimensions of interest by multiple types of stakeholders, such as users, operators, managers, providers and advocacy groups. LEAP utilises full-stack development, staging and production environments, composed of cutting-edge technologies such as, Git, Jenkins, Terraform, logic apps, Synapse, Docker containers, React.js, Tensorflow and .NET services and Python.

III. MULTI-CAMPUS UNIVERSITY SETTING

La Trobe University was founded in 1964, currently ranked in the top 400 of all three major world university rankings, and offers undergraduate, postgraduate, technical and executive education courses to a student load of approximately 39,000 and employs approximately 3250 staff with a total operating revenue of \$770 million[24]. The La Trobe University utilities network is spread across its multiple campuses in geographically distributed regions in the state of Victoria, Australia. The main metropolitan campus is La Trobe's Bundoora campus, which is spread over 235 hectares, it is the largest university campus in the southern hemisphere. The other four campuses are in regional locations, Bendigo, Shepparton, Albury-Wodonga and Mildura. All five campuses are in the state of Victoria which has a temperate oceanic climate, and four seasons: summer from December to February, autumn from March to May, winter from June to August and spring from September to November. The relevant physical characteristics of each campus are described below.

Bundoora Campus –located in Bundoora, 16 km north-east of Melbourne, the state capital of Victoria. The average minimum and maximum temperatures in Bundoora across the four seasons are tabulated in Table 1. Bundoora campus has a central heating system powered by gas that connects to all the buildings, while electricity powers all other amenities. This campus has a mix of old and modern buildings where several LED retrofit and solar installation projects have been carried out from 2019 to 2021. It also has 13x absorption chillers and 18x air cooled electric chillers.

TABLE I. AVERAGE SEASONAL TEMPERATURES (°C)

Campus	Summer		Autumn		Winter		Spring	
	Min	Max	Min	Max	Min	Max	Min	Max
Bundoora	13.6	26.2	10.1	20.5	5.9	14.1	8.9	19.9
Wodonga	14.5	30.5	9.1	22.5	3.8	13.8	8.2	21.6
Shepparton	14.5	30.7	8.9	22.5	3.6	14.1	7.9	22.2
Mildura	16.1	31.5	10.5	23.7	4.9	16.2	9.9	24.2
Bendigo	13.6	29.1	8.4	21.4	3	13.4	7	21

Bendigo Campus –located in Bendigo, a regional city, 150 kilometres north-west of Melbourne. See Table 1 for average minimum and maximum temperatures in Bendigo. The Bendigo campus has nine (9) Air Cooled Electric Chillers (2x1000kW, 6x630kW, 2x1000kW).

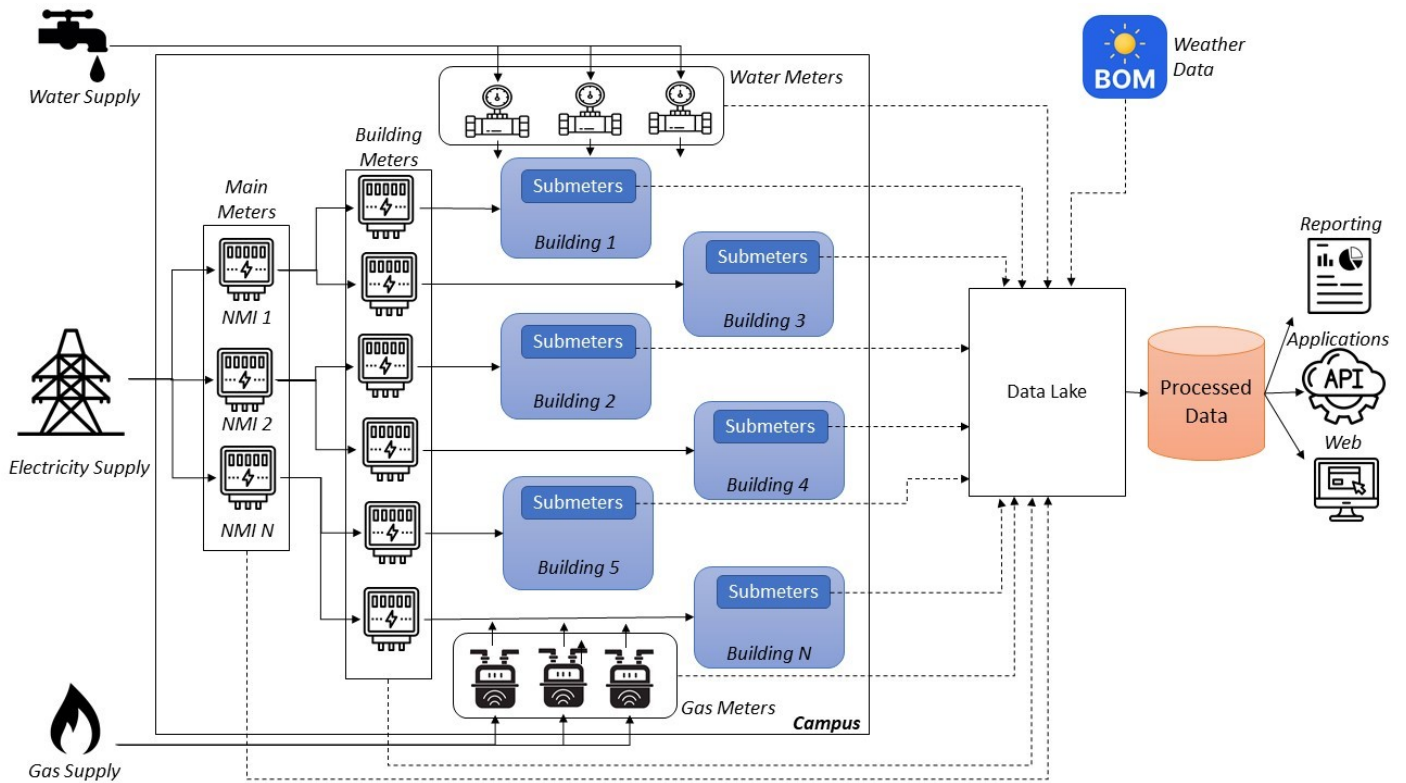


Fig.1. Overall architecture and data collection process

Shepparton Campus –located in Shepparton in northern Victoria, 181 kilometres north-northeast of Melbourne. See Table 1 for average minimum and maximum temperatures in Shepparton. It has a single 230kW Air Cooled Electric Chiller and 70kW Gas Chiller. Shepparton campus achieved its NetZero carbon emission certification from Climate Active in 2022 by going through a series of energy efficiency projects including Solar Installations and LED retrofits[25].

Albury-Wodonga Campus –located in the twin regional cities of Albury and Wodonga at the border of Victoria and New South Wales, Wodonga is 300 kilometres north-east of Melbourne. See Table 1 for average minimum and maximum temperatures in Wodonga. It has three (3) Air Cooled Chillers of 630kW cooling output.

Mildura Campus –located in Mildura, a regional city in north-west Victoria, 550 km from Melbourne. See Table 1 for average minimum and maximum temperatures in Mildura. This campus achieved NetZero carbon emission certification status from Climate Active in 2021

IV. THE UNICON DATASET

Fig. 1 illustrates the LEAP metering architecture which is represented as the data structure of the UNICON dataset and data collection approach. UNICON contains data from 2018 to 2021 inclusive, across all five campuses, leading up to a total of 14 National Meter Identifiers (NMIs), 71 university buildings, 10 submeter meters, 2 gas meter consumption data, and 5 water meters. The raw data is loaded into a loading zone hosted in a data lake through a streaming mechanism and thereafter pre-processed, augmented and stored in a primary data repository

which is hosted in a data lakehouse. Electricity data is grouped into three categories, primary, building and sub-meter as described below.

Primary meter data – This data is available for each NMI which provides electricity to the campuses in the university, this data is used to determine electricity consumption at a campus and university scope. The primary meter data is obtained through two sources; 1) the metering provider in NEM12 format 2) processed energy data from the energy retailer. Both 1) and 2) provide the same data, but it still useful as a source of validation.

Building data – Each building on campus has a separate smart meter installed on the building which is used to extract the data in CSV format and pushed into the data lake.

Submeter data – Each building has several subsystems such as lighting and mechanical panels. Smart meters are placed in these subsystems to capture the electricity consumption which is then pushed into the data lake with a unique identifier to identify the subsystem and the building

Natural gas is used as the main source of energy for the central heating systems used in the campuses. The gas meter readings are exposed through smart meters. Readings are hourly and are available at the end of each day, A gas day is 24 hours, commencing at 0600 Australian Eastern Standard Time.

Water data is collected from the metering infrastructure installed and maintained by the water service provider. This data is received in 15 min granularity, with submeters tracking the water disbursement at a building level.

The following subsections describe further characteristics of the dataset; 1) Building Types, 2) Weather Information, 3) Academic Calendar, 4) COVID-19 impact.

A. Building Types

Buildings in all campuses are used for different purposes and have varying energy consumption patterns and operational timings. Based on this primary purpose, we have grouped the buildings into the following seven categories. These categories are included as metadata to be used for various modelling and analysis requirements.

1. *Teaching Buildings*: Buildings used for lectures, workshop, tutorials, labs, other teaching activities, as well as all fundamental and applied research activities. Labs/research buildings also have more consistent load 24x7 due to low temp refrigeration and requirement for HVAC.
2. *Library Buildings*: buildings that provide library facilities and services, as well as large computer labs and study spaces for students. These buildings are typically open till midnight.
3. *Administrative buildings*: buildings used exclusively for university administration, such as the Finance, IT Services, HR and other admin functions. These used only during work hours, typically 8 am to 6 pm.
4. *Residential Buildings*: buildings used to provide accommodation facilities for students, includes shared spaces and private apartments
5. *Mixed-Use Buildings*: some buildings are used for mix of teaching, research and admin functions, they contain lecture rooms, office spaces, and study areas
6. *Sports Buildings*: These buildings are used for sports and recreational activities
7. *Other buildings*: Buildings that don't fall into any of the above categories are classified into this category, some examples are; greenhouses and animal house which are operational 24/7.

B. Weather Data

Weather data is strong correlated to utilities consumption and therefore vital for accurately predicting the consumption behaviours [26][27]. In the UNICON dataset, we can observe that the university network has a high energy demand during the summer periods compared to the winter. Hence capturing weather data is crucial information for any energy-related dataset. Weather data is publicly available from the Australian Bureau of Meteorology (BOM). Since the La Trobe university network is spread across multiple locations, we have used the closets weather station to gather the weather data. The weather records from the BOM contain several key parameters in 1-minute granularity and we have extracted temperature, humidity, wind speed, and wind direction data in this dataset. Table 1 shows weather station information.

TABLE II. WEATHER STATIONS

Campus	Weather Station	Station ID	Lat	Lon
Bundoora	Viewbank	086068	-37.74	145.10

Wodonga	Albury airport aws	072160	-36.07	146.95
Shepparton	Shepparton airport	081125	-36.43	145.39
Mildura	Mildura airport	076031	-34.24	142.09
Bendigo	Bendigo airport	081123	-36.74	144.33

The data is received at a two-speed latency with the most recent data being read through a JSON based service from the BOM which is refreshed every 10 minutes and contains 72 hours of data in 30-minute intervals. In addition to this, a daily CSV based export provides 1-minute data via File Transfer Protocol (FTP). The processing layer provides the weather data stream with the most recent data from the JSON data and then supplements the stream with the 1-minute data as and when the more granular data becomes available.

C. Academic Calendar

The academic calendar of the university shows working, non-working, semester days, and holidays with exam periods. We encode all this information into a CSV file which contains this information. The start of the first semester in the first quarter of the year reports the highest energy consumption period. It's because a large number of students are back at the university to attend special programs including orientation in this period and the university utilizes electricity to provide heating for the buildings and student accommodations. This information would be helpful for energy forecasting applications.

TABLE III. OVERVIEW OF DATA AVAILABILITY

Data	#Meters	Granularity	Campus	Time Range
NMIs	14	15m	All Campuses	2015-2022
Buildings	71	15m	All campuses	2018-2022
Submeters	9	5m	Bundoora	2019-2022
Gas	2	1h	Bundoora, Bendigo	2018-2022
Water	3	daily	Bundoora	2021-2022

D. COVID-19 Impact

COVID-19 pandemic forced global lockdowns and closed borders in early 2020 to curb the spread of the virus which resulted in low energy demand and emissions[28]. During the pandemic, Australia went through multiple lockdowns due to several COVID-19 outbreaks. Academic activities across universities in the world transform into online learning mode which changes energy consumption behaviours of the academic buildings. Hence analysing the influence of COVID-19 in the energy consumption of the buildings will be useful in understanding the current energy use and planning for new energy efficiency projects[29]. Many predictive models built prior to the COVID-19 are now no longer valid and it's required to build new models which reflect the new operating conditions. The UNICON dataset contains 2-3 years of pre-pandemic and close to 2 years of data during the pandemic which will be immensely helpful for energy researchers to analyse the pre- and

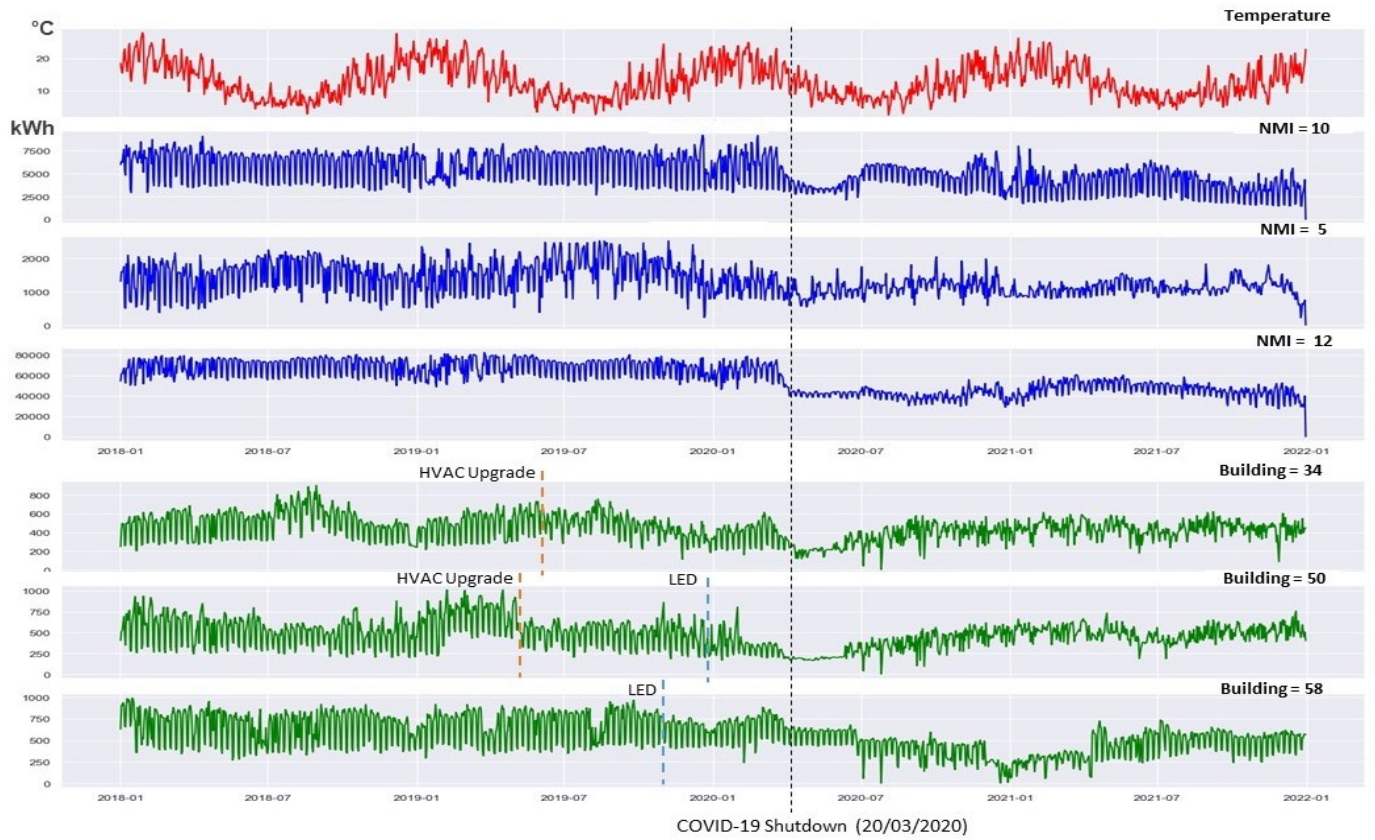


Fig. 2. Impact of multiple ECMs and COVID-19 shutdown across NMIs and Buildings

post-pandemic energy consumption patterns and build new algorithms. Effect of the COVID-19 on energy consumption of some NMIs and buildings are shown in Fig. 2.

V. DATA DICTIONARY

Energy data collected from each NMI, building, and sub-meters are stored in the form of CSVs. Table 4 and 5 contains detailed description about attributes of the collected data. The timestamps recorded in the data are Australian Eastern Standard Time. Each CSV file is described below.

- *campus_meta.csv* – This file contains information about each campus in the university network.
- *nmi_meta.csv* – Information about NMIs such as campus location and peak demand is listed in this file.
- *building_meta.csv* – This file contains meta information about buildings in each campus which include campus location, floor area and etc.
- *calender.csv* – University calendar for the data collection period is included in this file.
- *events.csv* – There are series of events happened at each building which include energy efficiency projects such as LED installation and HVAC system updates. This file contains the dates related to each event at building level.
- *nmi_consumption.csv* – Consumption data of NMIs are recorded in this file.

- *building_consumption.csv* – Consumption data of buildings are recorded in this file.
- *building_submeter_consumption.csv* – Consumption data of building sub-meters are recorded in this file.
- *gas_consumption.csv* – Gas consumption data of available campuses are recorded in this file.
- *water_consumption.csv* – Water consumption data of available campuses are recorded in this file.
- *weather_data.csv* – Weather data collected from respective weather station mentioned in Table 1 contain in this file.

TABLE IV. OVERVIEW OF META DATA

Category	Data field	Description	Unit	Data output
Campus	id	Unique identifier	-	campus_meta.csv
	name	Name of the campus	-	
	capacity	Student capacity	-	
NMI	id	Unique identifier	-	nmi_meta.csv
	campus_id	Campus location of NMI	-	
	peak_demand	Maximum peak demand	kVA	
Building	id	Unique Identifier	-	
	campus_id	Campus location of building	-	

	built_year	Year of construction	-	building_met a.csv
	category	Category based on usage	-	
	gross_floor_area	Gross floor area of the building	ft ²	
	room_area	Total room area	ft ²	
	capacity	Maximum user capacity	-	
Calender	date	Date		calender.csv
	is_holiday	Holiday or not		
	is_semester	Semester day or not		
	is_exam	Exam day or not		
Events	date	Date of the event		events.csv
	meter_id	Foreign key identifier to the building id		
	event_type	Type of the event		
	event_description	Description of the event		

TABLE V. OVERVIEW OF PRIMARY DATA

Category	Data field	Description	Unit	Data output
NMI Consumption Data	meter_id	Foreign key identifier to the NMI meter id	-	nmi_consumption.csv
	timestamp	Data recorded time	-	
	consumption	Energy consumption	kWh	
	demand_kW	Energy demand in kW	kW	
	demand_kVA	Energy demand in kVA	kVA	
Building Consumption Data	campus_id	Foreign key identifier to the campus id		building_consumption.csv
	meter_id	Foreign key identifier to the building id	-	
	timestamp	Data recorded time	-	
	consumption	Energy consumption	kWh	
Building Submeter Consumption Data	id	Unique identifier		building_submeter_consumption.csv
	building_id	Foreign key identifier to the building id	-	
	timestamp	Data recorded time	-	
	consumption	Energy consumption	kWh	
	current	Current consumption	A	
	voltage	Voltage	V	
	power	Power Consumption	kW	
	power_factor	Power Factor		
Gas Consumption Data	campus_id	Foreign key identifier to the campus id	-	gas_consumption.csv
	timestamp	Recorded timestamp	-	
	consumption	Gas consumption	kJ	

Water Consumption Data	campus_id	Foreign key identifier to the campus id	-	water_consumption.csv
	meter_id	Unique meter identifier		
	timestamp	Recorded timestamp	-	
	consumption	Water consumption	kL	
Weather Data	campus_id	Foreign key identifier to the campus id	-	weather_data.csv
	timestamp	Recorded timestamp	-	
	apparent_temperature	Apparent Temperature	°C	
	air_temperature	Air Temperature	°C	
	dew_point_temperature	Dew Point Temperature	°C	
	relative_humidity	Relative Humidity	-	
	wind_speed	Wind Speed	m/s	
	wind_direction	Wind Direction angle w.r.t. to North	°	

VI. TECHNICAL VALIDATION

As the final contribution of this article, we conducted technical validation of the dataset for consumption trends, baseline modelling and forecasting. These are presented in the following subsections.

A. Consumption trends

According to the average temperature and the energy consumption patterns in Fig. 2, we can observe that temperature has both positive and negative impact for the electricity consumption which is based on the NMI or building. The campuses across the university network were shut down due to the COVID-19 pandemic starting 20/03/2020. Fig. 2 shows the

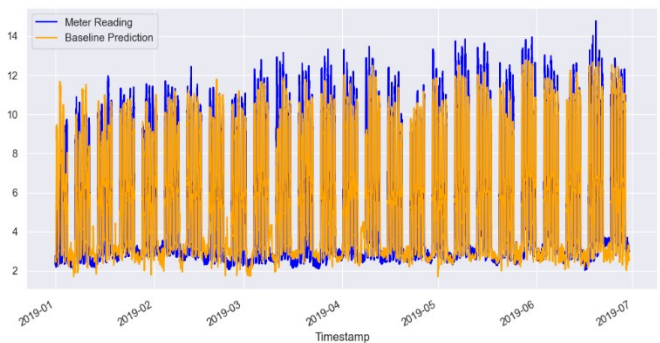


Fig. 3. XGBoost baseline prediction for #33 building

reduction in electricity consumption due the COVID-19 across NMIs and buildings. There were several energy efficiency projects carried out in buildings of the university. As shown in the Fig. 2, building #34 subjected to a HVAC system tuning on 10/06/2019 which reduced the energy consumption. There were two energy efficiency projects carried out in the building #50. One project was a HVAC system tuning which happened on 23/05/2019. This building had old light bulbs which completely replaced by LED lights to reduce the electricity consumption. This project was carried on 13/12/2019. LEDs of building #58 also replaced by LED light bulbs on 30/10/2019. Impact of these

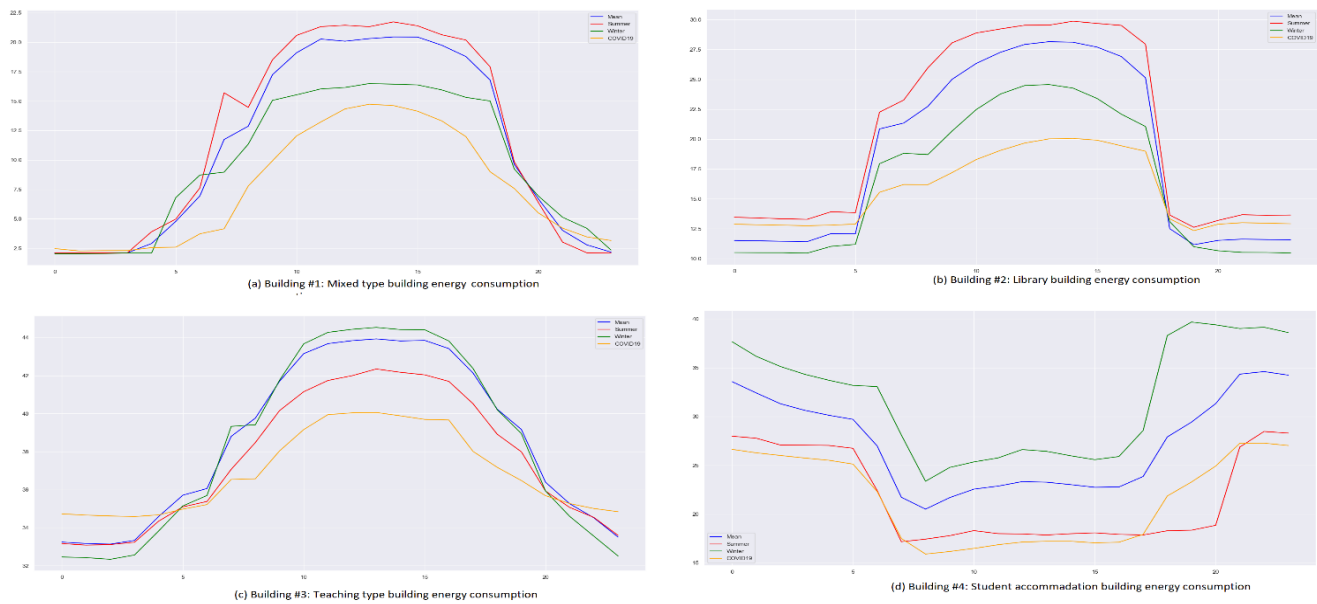


Fig. 4: Daily average electricity consumption comparison with Summer, Winter and COVID19 periods

projects can be seen in Fig. 2. Information including dates about several events and energy efficiency projects which change the electricity consumption pattern of building are contain in the events data file.

Dataset contains electricity consumption data of 71 buildings. Most of the buildings in the university network show a high energy consumption during the summer period compared to the winter period. But not all the buildings follow a similar pattern. Fig. 4 shows the average daily energy consumption for a selected set of buildings which shows the high energy consumption in summer and low energy consumption in winter. Fig. 4 also indicates the impact of the average energy consumption during the COVID-19 period. Accommodation building shows an inverse energy consumption pattern compared to other buildings. It also shows the impact of the availability of sunlight during the day in each season and an increase in energy consumption in summer. This type of benchmarking enables LEAP to identify opportunities to update building operating parameters to try and match best in class buildings

B. Baseline Modeling

Baseline modelling predicts the standard energy consumption behaviour of a building based on its historical pattern. Energy baseline is a critical requirement to estimate the energy savings of a building result from ECM project. Fig.3 shows baseline

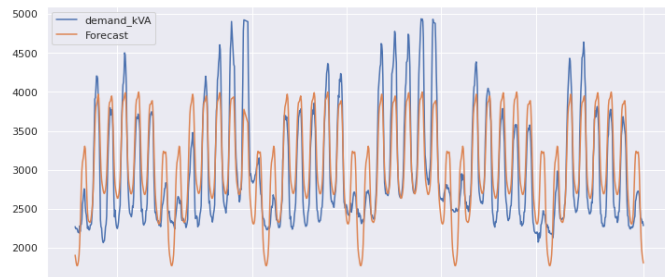


Fig. 5. Forecast of #12 NMI electricity demand using FBProphet

energy predictions from a XGBoost model which include weather variables as well. This model is trained using data from 2018 and predicted for the first six months of 2019. It has a R2 score of 0.91 and Root Mean Square Error (RMSE) value of 1.08.

C. Forecasting

Building energy forecasting models is one of the major use cases of the UNICON dataset. Fig. 5 shows the output of a univariate FBProphet forecasting model for demand forecasting of NMI which trained on data from 01/01/2019 to 21/01/2020 data and forecasted for 6 weeks from 22/01/2020 which has a Mean Absolute Percentage Error (MAPE) value of 11.1% and RMSE value of 380.71.

VII. DATASET AVAILABILITY

All the energy data are stored in CSV format. Hence, any standard CSV reading library can be used to parse the data. The dataset can be downloaded from the following link, which also contains a Jupyter notebook to get started <https://github.com/CDAC-lab/UNICON>. Please cite this article in your work.

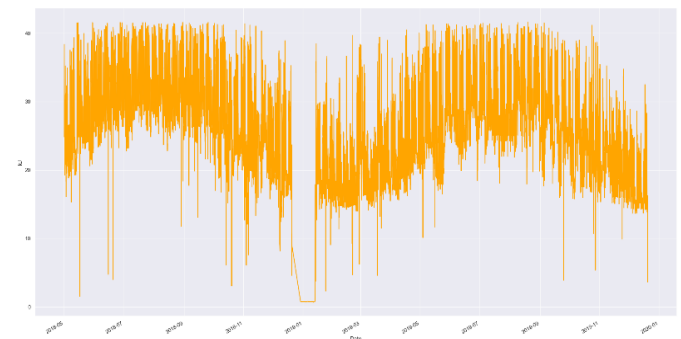


Fig. 6. Gas consumption of Bundoora Campus

VIII. CONCLUSION

In this article, we have introduced UNICON, a large-scale open dataset of electricity, gas and water consumption in a large multi-campus university setting. This dataset is released for public use as part of La Trobe University's commitment to net zero carbon emissions by 2029. The UNICON dataset contains consumption data for La Trobe's five campuses in geographically distributed regions, across four years, 2018-2021 inclusive. We have documented the characteristics and behaviours of the dataset, a data dictionary, as well as conducted a technical validation of consumption trends, baseline modelling and forecasting as contributions of this article. To the best of our knowledge, this is the first large-scale, comprehensive, open dataset for the three main utilities, electricity, gas, and water consumption in a multi-campus university setting.

REFERENCES

- [1] D. Larcher and J. Tarascon, "Towards greener and more sustainable batteries for electrical energy storage," *Nat. Publ. Gr.*, no. November, pp. 1–11, 2014, doi: 10.1038/nchem.2085.
- [2] Y. Himeur, A. Alsalemi, F. Bensaali, and A. Amira, "Energy & Buildings Building power consumption datasets: Survey, taxonomy and future directions," *Energy Build.*, vol. 227, p. 110404, 2020, doi: 10.1016/j.enbuild.2020.110404.
- [3] X. Liang, T. Hong, and G. Q. Shen, "Improving the accuracy of energy baseline models for commercial buildings with occupancy data," *Appl. Energy*, vol. 179, pp. 247–260, 2016, doi: 10.1016/j.apenergy.2016.06.141.
- [4] J. Chou and D. Bui, "Modeling heating and cooling loads by artificial intelligence for energy-efficient building design," *Energy Build.*, vol. 82, no. 2014, pp. 437–446, 2020, doi: 10.1016/j.enbuild.2014.07.036.
- [5] S. Makonin, B. Ellert, I. V. Bajic, and F. Popowich, "Data Descriptor: Electricity, water, and natural gas consumption of a residential house in Canada from 2012 to 2014," *Sci. Data*, pp. 1–12, 2016, doi: https://doi.org/10.1038/sdata.2016.37.
- [6] EPA. *Public law 109-58: 109th congress: an act to ensure jobs for our future with secure, affordable, and reliable energy*. 2005.
- [7] European Parliament and Council, *Directive 2012/27/EU of the European parliament and of the council of 25 October 2012 on energy efficiency*. 2012.
- [8] T. Walter, P. N. Price, and M. D. Sohn, "Uncertainty estimation improves energy measurement and verification procedures," *Appl. Energy*, vol. 130, pp. 230–236, 2014, doi: 10.1016/j.apenergy.2014.05.030.
- [9] C. V. Gallagher, K. Leahy, P. O'Donovan, K. Bruton, and D. T. J. O'Sullivan, "Development and application of a machine learning supported methodology for measurement and verification (M&V) 2.0," *Energy Build.*, vol. 167, no. 2018, pp. 8–22, 2018, doi: 10.1016/j.enbuild.2018.02.023.
- [10] I. H. Sarker, "Machine Learning: Algorithms, Real - World Applications and Research Directions," *SN Comput. Sci.*, vol. 2, no. 3, pp. 1–21, 2021, doi: 10.1007/s42979-021-00592-x.
- [11] T. Kempitaya, S. Sierla, D. De Silva, and M. Yli-ojanperä, "An Artificial Intelligence framework for bidding optimization with uncertainty in multiple frequency reserve markets," *Appl. Energy*, vol. 280, no. September, p. 115918, 2020, doi: 10.1016/j.apenergy.2020.115918.
- [12] D. A. C. Narciso and F. G. Martins, "Application of machine learning tools for energy efficiency in industry: A review," *Energy Reports*, vol. 6, pp. 1181–1199, 2020, doi: 10.1016/j.egy.2020.04.035.
- [13] N. J. Johannesen, M. Kolhe, and M. Goodwin, "Relative evaluation of regression tools for urban area electrical energy demand forecasting," *J. Clean. Prod.*, vol. 218, pp. 555–564, 2019, doi: 10.1016/j.jclepro.2019.01.108.
- [14] C. Klemenjak, A. Reinhardt, L. Pereira, S. Makonin, and M. Bergés, "Electricity Consumption Data Sets: Pitfalls and Opportunities," *Assoc. Comput. Mach.*, pp. 159–162, 2019, doi: https://doi.org/10.1145/3360322.3360867.
- [15] O. Parson *et al.*, "Dataport and NILMTK: A building data set designed for non-intrusive load monitoring," in *2015 IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, 2015, pp. 210–214, doi: 10.1109/GlobalSIP.2015.7418187.
- [16] K. Anderson, A. Ocleanu, D. Benitez, D. Carlson, A. Rowe, and M. Berges, "BLUED: A fully labeled public dataset for event-based non-intrusive load monitoring research," in *Proceedings of the 2nd KDD workshop on data mining applications in sustainability (SustKDD)*, 2012, vol. 7, pp. 1–5.
- [17] S. Makonin, "HUE: The hourly usage of energy dataset for buildings in British Columbia," *Data Br.*, vol. 23, p. 103744, 2019, doi: https://doi.org/10.1016/j.dib.2019.103744.
- [18] S. Makonin, Z. J. Wang, and C. Tumpach, "RAE: The rainforest automation energy dataset for smart grid meter data analysis," *data*, vol. 3, no. 1, p. 8, 2018.
- [19] T. Kriebchaumer and H.-A. Jacobsen, "BLOND, a building-level office environment dataset of typical electrical appliances," *Sci. data*, vol. 5, no. 1, pp. 1–14, 2018.
- [20] T. Picon *et al.*, "COOLL: Controlled on/off loads library, a public dataset of high-sampled electrical signals for appliance identification," *arXiv Prepr. arXiv1611.05803*, 2016.
- [21] N. Batra, M. Gulati, A. Singh, and M. B. Srivastava, "It's Different: Insights into Home Energy Consumption in India," in *Proceedings of the 5th ACM Workshop on Embedded Systems For Energy-Efficient Buildings*, 2013, pp. 1–8, doi: 10.1145/2528282.2528293.
- [22] C. Shin, E. Lee, J. Han, J. Yim, W. Rhee, and H. Lee, "The ENERTALK dataset, 15 Hz electricity consumption data from 22 houses in Korea," *Sci. data*, vol. 6, no. 1, p. 193, Oct. 2019, doi: 10.1038/s41597-019-0212-5.
- [23] H. Rashid, P. Singh, and A. Singh, "Data Descriptor: I-BLEND, a campus-scale commercial and residential buildings electrical energy dataset," *Nat. Publ. Gr.*, pp. 1–12, 2019, doi: 10.1038/sdata.2019.15.
- [24] "Facts and figures." [Online]. Available: https://www.latrobe.edu.au/about/at-a-glance/facts-figures.
- [25] "La Trobe meets Net Zero regional targets." [Online]. Available: https://www.latrobe.edu.au/news/articles/2022/release/la-trobe-meets-net-zero-regional-targets. [Accessed: 01-Jul-2022].
- [26] Y. Geng, W. Ji, B. Lin, J. Hong, and Y. Zhu, "Energy & Buildings Building energy performance diagnosis using energy bills and weather data," *Energy Build.*, vol. 172, pp. 181–191, 2018, doi: 10.1016/j.enbuild.2018.04.047.
- [27] J. Koči, V. Koči, J. Maděra, and R. Černý, "Effect of applied weather data sets in simulation of building energy demands: Comparison of design years with recent weather data," *Renew. Sustain. Energy Rev.*, vol. 100, no. October 2018, pp. 22–32, 2019, doi: 10.1016/j.rser.2018.10.022.
- [28] C. Bertram *et al.*, "COVID-19-induced low power demand and market forces starkly reduce CO₂ emissions," *Nat. Clim. Chang.*, vol. 11, no. March, 2021, doi: 10.1038/s41558-021-00987-x.
- [29] N. Luo *et al.*, "A three-year dataset supporting research on building energy management and occupancy analytics," *Sci. Data*, pp. 1–15, 2022, doi: 10.1038/s41597-022-01257-x.